

GAUTAM KHOSLA

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SUMMARY

Computer Engineering co-op student who builds for the failure case and reports it when it happens: hard real-time C++20 on QNX with no network in the safety loop, agent infrastructure that refuses to boot rather than run unsafely, and a pre-registered study published with its negative result intact. Source, write-ups and live demos at gautamkhosla.com.

SKILLS

Systems: C++20, Rust, Go, QNX 8.0 RTOS, SCHED_FIFO real-time scheduling, POSIX message passing, eBPF, gRPC/mTLS, CMake cross-compilation, Linux, chaos and fault-injection testing

Product: TypeScript, React, Next.js 14, Node.js/Express, FastAPI, REST, WebSockets, Server-Sent Events, Tailwind

Data & Infra: PostgreSQL (Row-Level Security, Prisma), Supabase, MongoDB, Redis, Snowflake, Docker, Kubernetes/Helm, Terraform, NATS, BullMQ, Turborepo/pnpm, Stripe

Applied AI: Multi-agent orchestration, RAG and vector retrieval, model routing by latency/cost, evaluation harnesses and guardrails (Anthropic, Gemini, Groq, Cerebras)

Statistics: Pre-registration, censored (Tobit) regression, cross-validated interval estimation, bootstrap confidence intervals, variance decomposition, selection-bias checks

PROJECTS

AEGIS — Edge-AI Worksite Safety on QNX RTOS · cuHacking 7, 2026 · C++20 · QNX 8.0 · Raspberry Pi 5 · TFLite · Next.js · MongoDB Atlas

github.com/GautamTalksDev/AEGIS · *architecture, failure-mode table and latency budget in docs/ARCHITECTURE.md*

- Drew an explicit safety boundary and enforced it in code: only `aegis_safety` writes GPIO, only `aegis_uplink` opens a socket. Gateway, database, Gemini and ElevenLabs can all be down and the relay still de-energises. Validated end to end on QNX-on-Pi hardware with a live camera, not in simulation.
- Split the edge into four QNX processes with distinct SCHED_FIFO priorities and CPU affinity (safety 30/core 0, capture 20, vision 15, uplink 10) and chose a different IPC primitive per link by what each one needs: a shared-memory seqlock ring for frames (freshness over completeness), synchronous `MsgSend` rendezvous for detections (backpressure instead of unbounded queues), and a lock-free SPSC ring for telemetry (droppable, never blocks safety).
- Made the safety process observable to itself: separate SHM heartbeat words let a 50 ms timer pulse detect capture or vision death even when `MsgReceive` never wakes, driving a latching `FAIL_SAFE` that requires 5 s of continuous health to clear. Documented a 13-row failure-mode table in which only total loss of power affects worker safety.
- Instrumented photon-to-alarm latency in-band against a sub-100 ms design budget: capture timestamp subtracted from monotonic time at the GPIO write, kept in a 256-sample ring, reported as p99 and max on every violation rather than as a mean.
- Guaranteed zero telemetry loss across network partitions with a CRC32 write-ahead log written before any HTTP attempt, a cursor that advances only on a gateway 2xx, and idempotent cloud ingest on event ID so crash-replay double-sends are harmless. Emergency voice alerts are baked into the device image at build time, on the principle that an alarm requiring an API call is not an alarm.

label-to-lab — Pre-Registered Test of Label-Based Contaminant Prediction · 2026 · Python · Tobit regression · FDA and Open Food Facts data

github.com/GautamTalksDev/label-to-lab · MIT, citable, pre-registration and test-set hash committed before modelling

- Pre-committed the threshold, metrics and failure rule, hashed the test wave before seeing it, then published the negative result: 9.8% RMSE reduction against a pre-registered 20% bar. No model was refit after unsealing, and an unused baseline stub was left empty rather than back-filled.
- Caught a bug in my own pipeline that a results-only writeup would have buried: a fuzzy matcher returning an implausible 97.4% match rate because `token_set_ratio` scores 100 on subsets, collapsing 991 products onto 436 records. Built a distinct-ID-count diagnostic to detect it, rebuilt the matcher, and cut collapse to 1.08x.
- Bounded what the method could ever achieve before blaming the model: pooled within-product replicate SD of 0.223 log10 caps any label-based predictor near 50% reduction, and an apparent reformulation signal proved to be category composition change, with only 3.5% product overlap between waves.

Cadence — Production Inventory & Rental System · Canadian Musicians Cooperative, 2026 · TypeScript · Next.js · Prisma · PostgreSQL · Stripe

- Sole engineer from requirements gathering with non-technical staff to a deployed, documented system: role-based access control, Stripe billing, full English/French support via next-intl, plus explicit build, defer and buy calls under a fixed timeline. Delivered and handed off; the cooperative later changed direction on tooling.

HyperShift — Autonomous Cloud Orchestration · 2025 · Team · TypeScript · Next.js · Turborepo · BullMQ · Zod · PostgreSQL

- Turned plain-English intent into deployed, monitored infrastructure through a five-stage agent pipeline on durable BullMQ queues, with approval gates so nothing reaches production without a human sign-off.
- Shipped the full developer surface (CLI, workspace isolation, usage metering, per-run observability traces) and enforced shared Zod contracts across every service boundary so integration breakage failed at compile time rather than in production.

AeroGuard — SOC Incident Triage · ★ MLH Best Use of Snowflake, HackConcordia 2025 · Python · FastAPI · React · MongoDB · Snowflake

- Dual-store architecture separating live operational reads (MongoDB) from analytical processing (Snowflake) so analytics never contend with real-time state. That decision won the award. Shipped in 24 hours with an automated E2E suite.

Also: MetaShift, multi-tenant SaaS with PostgreSQL RLS, JWT and API-key auth, real-time conflict resolution over WebSockets, metering and billing · Chronos Cloud, technical lead of 4, 14 live APIs over NATS with graceful upstream-outage degradation · NeuroDesk · Omni-Context OS, Top 9 at Backboard.io × McHacks.

WORK EXPERIENCE

Data Processor & Software Developer — Canadian Musicians Cooperative · Jul 2026 – Aug 2026

- Built Cadence (above) solo; audited and repaired data-integrity failures across cooperative tooling; automated recurring workflows with Apps Script, Slack webhooks, and scheduled triggers.

Financial Sales Representative — Kognitive Sales Solutions × Canadian Tire · Oct 2025 – Jan 2026

- Explained regulated financial products under strict compliance at high volume: daily practice in making complicated things understandable to people who did not ask for the details.

Class Representative, GNG1105 — uOttawa Engineering Students' Society · Sep 2024 – May 2025

- Primary liaison between roughly 100 students and faculty; gathered course feedback, identified recurring issues, and escalated them through departmental meetings until resolved.

EDUCATION

B.A.Sc. Computer Engineering (Co-op) — University of Ottawa · Sep 2024 – May 2029 (expected) · Ottawa, ON

Data structures & algorithms, databases, software engineering, computer architecture, discrete mathematics, probability & statistics.